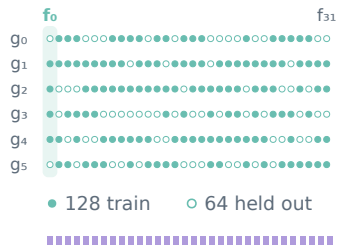
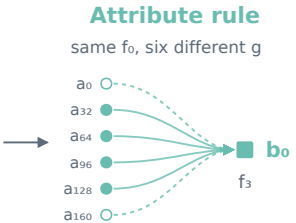
# A Attribute-defined object prediction

## 01 Assign categorical attributes




## 02 Create two matched relations




## 03 Repeat & shuffle sentences

**Attributes · all 224 entities** ×128

x $a_0$ $f_0$ $g_0$ x

x $b_0$ $f_3$ $g_0$ x

**Attribute rule · 128 triples** ×16

x $a_{32}$ $r_0$ $b_0$ x

**Random control · 128 triples** ×16

x $a_{32}$ $r_1$ $b_{15}$ x

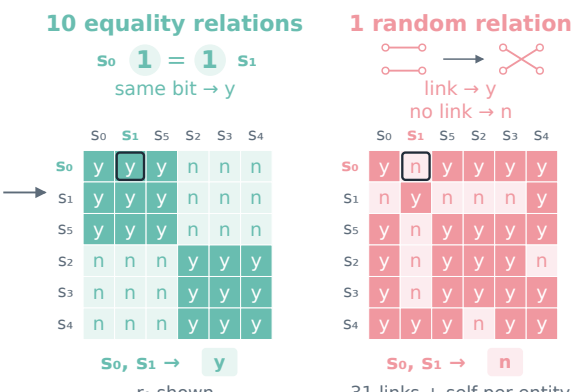
# B Attribute equality & symmetric lookup

64 entities → binary pair labels

## 01 Assign each entity 10 bits

6 of 64 entities shown

## 02 Label every entity pair




## 03 Write training sentences

**Attributes · forward + inverse**

$S_0$ $attr_0$ → 1

$attr_0$ 1 → $S_0$

**Same attribute**

$r_0$ $S_0$ $S_1$ → y

**Random lookup**

$r_{10}$ $S_0$ $S_1$ → n

Joint training after the equality warm-up

10 equality relations · 1 random relation

Random: 10× each equality relation